Introduction to Ratios

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Introduction to ratio

What is a ratio?

- A **ratio** is a way of comparing one **part** of a **whole** to another
  - **Ratios** are used to compare **one part** to **another part**
    - **How many** parts that make the whole will depend on the question

What do ratios look like?

- Ratios involve **two or three different numbers** separated using a **colon**
  - E.g. 2 : 5, 3 : 1, 4 : 2 : 3
- In all ratio questions, who or what is **mentioned first** in the question, will be associated with the **first part of the ratio**
  - E.g. The cake recipe with flour and butter in the ratio 2 : 1
    - 'Flour' is associated with '2' and 'butter' is associated with '1'
- The numbers in a ratio tell us, for **each quantity** involved, its **proportion of the whole**
  - In the ratio **4 : 3**
    - The **first quantity** comprises 4 parts (of the whole)
    - The **second quantity** comprises 3 parts (of the whole)
    - In total, the **whole** is made up of 4 + 3 = 7 parts
  - In the ratio **2 : 5 : 3**
    - The **first quantity** comprises 2 parts (of the whole)
    - The **second quantity** comprises 5 parts (of the whole)
    - The **third quantity** comprises 3 parts (of the whole)
    - In total, the **whole** is made up of 2 + 5 + 3 = 10 parts

How are ratios and fractions linked?

- Fractions and ratios are closely linked and are used to compare a **part of a whole**
  - A **fraction** compares **a part to the whole**

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

slices, and the second person receives 3 slices
    - As a **fraction** (of the whole)
        - The first person receives $\frac{5}{8}$ of the pizza
        - The second person receives $\frac{3}{8}$ of the pizza

- The **ratio** of slices of the first person to the second person is 5 : 3
    - The 8 does not appear in the ratio but is obtained by adding the parts together (5 + 3 = 8)

- Unlike ratios, **fractions** can be converted/expressed as **percentages** or **decimals**

Examiner Tips and Tricks

Start by **writing down** any information given in words as **values**, and use **abbreviations** to refer to them.

This will make it easier to look back at the information later.

E.g. For a flour and butter cake mix, where twice as much flour is needed as butter, you could write:

$$\begin{array}{c} \text{F : B} \\ 2 : 1 \end{array}$$

Worked Example

A pot of money is shared between three friends, Dave, John and Mary.
Dave receives £450, John receives £200 and Mary receives £350.

(a) Find the total amount of money in the pot.

Answer:

Add up the three separate amounts

$$450 + 200 + 350 = 1000$$

£1000

(b) Write down the ratio of money received by Dave, John and Mary.
(There is no need to simplify the ratio.)

Answer:

Be careful with the order
Dave gets mentioned first, so 450 will be the first part of the ratio, then John and finally Mary

450 : 200 : 350

Answer:

Fractions are compared to the whole, so this will be 'Mary's money' "out of" 'total money'

$$\frac{350}{1000}$$

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